

Saksham Agarwal

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

B.Tech CSE student at VIT Bhopal (CGPA: 8.46/10, graduating 2027) with hands-on expertise in full-stack web development, applied ML/AI systems, and cybersecurity. Architect of production-grade systems spanning a multi-modal deepfake forensics platform with explainable AI, an LLM-powered natural-language app compiler, a real-time collaborative canvas engine, and a hybrid malware detection engine. Proficient across the Python and JavaScript ecosystems; experienced with FastAPI, React/Next.js, and Google Cloud Platform.

EDUCATION

VIT Bhopal University — B.Tech, Computer Science Engineering 2023 – 2027

- CGPA: 8.46 / 10.0

St. Anthony's Senior Secondary School — CBSE (Science Stream) 2021 – 2022

TECHNICAL SKILLS

- Programming Languages:** Python, Java, C++, JavaScript, TypeScript, C
- Web Development:** React.js, Next.js, FastAPI, Flask, Node.js, HTML, CSS, Bootstrap, RESTful APIs
- Machine Learning & AI:** PyTorch, scikit-learn, TensorFlow, Computer Vision (OpenCV, MediaPipe), Explainable AI (Grad-CAM, SHAP), Generative AI (LLM Pipelines/API), OCR (Tesseract)
- Database & Cloud:** SQL, Firebase, Appwrite, Google Cloud Platform, Cloud Architecture
- Tools & Technologies:** Git, GitHub, Docker, CI/CD, YARA, Jupyter Notebook, Linux

FEATURED PROJECTS

Deepfake Forensics & Explainable AI Platform | *Python, PyTorch, FastAPI, React, OpenCV, MediaPipe, EfficientNet-B4* [Code →](#)

- Architected a production-grade, multi-modal deepfake forensics engine fusing 15 detection signals — an EfficientNet-B4 visual classifier with Grad-CAM/SHAP explainability, SyncNet-based audio-visual lip-sync analysis, and rPPG/frequency-domain forensics — enabling robust detection across video, image, and audio modalities.
- Integrated a FastAPI backend with a React/Vite frontend that auto-generates annotated PDF evidence reports for each analyzed artifact.

NL App Compiler | *Node.js, JavaScript, Gemini API* [Code →](#)

- Designed a 4-stage, compiler-style LLM pipeline — intent parsing, UI/UX design generation, database schema synthesis, and cross-layer refinement — transforming natural-language prompts into validated, deployable web applications via chained Gemini API calls.

NexusBoard | *React 18, Tailwind CSS, WebSockets, Canvas API* [Code →](#)

- Built a real-time collaborative infinite canvas and whiteboard engine with live multi-user sync over WebSockets and high-performance freehand drawing on the Canvas API.

DocPilot – AI-Powered Clinical Management Platform | *React, TypeScript, Firebase, Appwrite, Gemini AI* [Code →](#)

- Engineered a full-stack, role-based clinical platform for doctors and patients, with Gemini AI as an intelligent consultation scribe that auto-generates structured clinical notes and visit summaries, backed by secure Firebase multi-role auth and real-time Appwrite data sync.

Malware Detector | *Python, YARA, Security Forensics* [Code →](#)

- Built a malware inspection engine combining custom YARA signature rules with heuristic behavioral analysis — performing static analysis of PE headers, entropy metrics, and behavioral indicators — to flag malicious file patterns across multiple malware families.

HACKATHONS & COMPETITIONS

- TCS CodeVita** — Global Rank 6735 of 20,540 — Secured Global Rank 6735 in Round 1.
- Gridlock Hackathon 2.0 (Flipkart)** — Round 2 Qualifier — Qualified solo in Flipkart's national ML hackathon, building a traffic-demand prediction model with 93.94% accuracy on HackerEarth.

PROFESSIONAL CERTIFICATIONS

Generative AI Professional Certificate (IBM, 2025) • AI & Edge Computing for Industry Applications (I&T EduTech, 2025) • Bits and Bytes of Computer Networking (Google, 2025) • [Google Cloud Skills Boost \(2025\)](#) • Introduction to Machine Learning (NPTEL–IIT Kanpur, 2025) • Marketing Analytics (NPTEL–IIT Kanpur, 2026)

Verify remaining credentials: [LinkedIn – Licenses & Certifications](#)

VOLUNTEERING & LEADERSHIP

Design Team Lead, Fintech Club — VIT Bhopal 2025 – Present

- Spearheaded the club's end-to-end visual identity — social media assets, event collateral, and digital campaigns — driving consistent brand presence across all activities.